FUN-3.E.1 The chain rule and definition of an inverse function can be used to find the derivative of an inverse function, provided the derivative exists.

3-3 Differentiating Inverse Functions

Get Ready...

Find $f^{-1}(x)$ if $f(x) = \sqrt{10 - 3x}$

$$y = \sqrt{10-3x}$$

$$x = \sqrt{10-3y}$$

$$x^2 = 10-3y$$

$$\frac{x^2-10}{-3} = \frac{-3y}{-3}$$

$$\frac{x^2-10}{-3} = y$$

$$-\frac{x^2}{3} + \frac{10}{3} = y \qquad f^{-1}(x) = \frac{-x^2}{3} + \frac{10}{3}$$

In this lesson, we will explore three different ways to find the derivative of an inverse function. Once we look at these different approaches, the goal is for you to understand how derivatives of inverses work regardless of what representation you are given in the problem!

The following methods can be used to evaluate the derivative of the inverse of a function at a point:

Method #1	
Use when: You are given an equation for $y = f(x)$ and can find the equation of the inverse	
Steps	**Notes**
1. Find the inverse function	Find inverse by interchanging x and y—then solve for y
2. Take derivative of the "new" y	This will be the derivative of the inverse function
3. Plug in the given point	

Example 1: Let f and g be inverse functions where $f^{-1}(x) = g(x)$. If $f(x) = x^3 - 1$, find $g'(7)$.

$$g'(7) = (f^{-1})'(7)$$
$$= \frac{1}{3}(7+1)^{-2/3}$$
$$= \frac{1}{3} \cdot \frac{1}{8^{2/3}}$$
$$= \frac{1}{3} \cdot \frac{1}{4}$$
$$= \frac{1}{12}$$

$$y = x^3 - 1$$
$$x = y^3 - 1$$
$$x + 1 = y^3$$
$$\sqrt[3]{x+1} = y$$
$$f^{-1}(x) = \sqrt[3]{x+1}$$
$$= (x+1)^{1/3}$$

$$(f^{-1})'(x) = \frac{1}{3}(x+1)^{-2/3} \cdot 1$$

Example 2: If $f(x) = e^{2x}$, find the derivative of $f^{-1}(x)$ at $x = 3$.

$y = e^{2x}$

$x = e^{2y}$

$\ln x = \ln e^{2y}$

$\ln x = 2y \cdot \ln e$

$\ln x = 2y$

$\frac{\ln x}{2} = y$

$f^{-1}(x) = \frac{\ln x}{2} = \frac{1}{2} \cdot \ln x$

$(f^{-1})'(x) = \frac{1}{2} \cdot \frac{1}{x}$

$(f^{-1})'(3) = \frac{1}{2} \cdot \frac{1}{3} = \frac{1}{6}$

Method #2	
Use when: You are given an equation for $y = f(x)$ but finding the inverse is difficult or impossible	
Steps	**Notes**
1. Find the inverse function	Find inverse by interchanging x and y—**DON'T** solve for y
2. Differentiate both sides of the new relation	This will require implicit differentiation
3. Solve for $\frac{dy}{dx}$	This expression will be in terms of y
4. Plug in the needed y −value	You may need to solve for y in the initial equation first

~~**Note: If you only need to find the derivative of the inverse, you will need to replace y with the original function.~~

Example 3: Find the derivative of the inverse function of $f(x) = x^3 + 2x + 1$ at $x = 1$.

$$y = x^3 + 2x + 1$$

$$x = y^3 + 2y + 1$$

find y when x=1 in inverse function

$$1 = y^3 + 2y + 1$$

$$-1 \qquad\qquad -1$$

$$0 = y^3 + 2y$$

$$0 = y(y^2 + 2)$$

$$y = 0 \qquad y^2 + 2 = 0$$

$$\cancel{y^2 = -2}$$

no sol

$$1 = 3y^2 \frac{dy}{dx} + 2\frac{dy}{dx}$$

$$1 = \frac{dy}{dx}(3y^2 + 2)$$

$$\frac{dy}{dx} = \frac{1}{3y^2 + 2}$$

$$\left.\frac{dy}{dx}\right|_{x=1} = \left.\frac{dy}{dx}\right|_{y=0} = \frac{1}{0+2} = \frac{1}{2}$$

Method #3
The Derivative of an Inverse Function

Let f be a function that is differentiable. If f has an inverse function f^{-1}, then f^{-1} is differentiable at any x for which $f'(f^{-1}(x)) \neq 0$. Then,

$$(f^{-1})'(x) = \frac{1}{f'(f^{-1}(x))}$$

−OR −

Let $g = f^{-1}$, and let $g(x) = y$. Then, $g'(x) = \dfrac{1}{f'(y)}$

***Note: This is not the best method for finding the derivative of the inverse but is great for finding the derivative of the inverse of the function at a point. I GUARANTEE this will be on the AP exam, so **MEMORIZE THIS FORMULA!**

Example 4: Let f be the function defined by $f(x) = x^3 + 2x^2 + 1$. Let $g(x) = f^{-1}(x)$, where $g(2) = -1$. Find the value of $g'(2)$.

$$g'(2) = (f^{-1})'(2)$$

$$= \frac{1}{f'(f^{-1}(2))}$$

$$= \frac{1}{f'(-1)}$$

$$= \frac{1}{-1} = \boxed{-1}$$

$f^{-1}(2) = ? \longleftrightarrow f(?) = 2$

$g(2) = f^{-1}(2) = -1$

$f'(x) = 3x^2 + 4x$

$f'(-1) = 3 - 4 = -1$

Example 5: If $f(x) = 3x^3 + 1$ and g is the inverse function of f, what is the value of $g'(25)$?

$$g'(25) = (f^{-1})'(25)$$

$$= \frac{1}{f'(f^{-1}(25))}$$

$$= \frac{1}{f'(2)}$$

$$= \boxed{\frac{1}{36}}$$

$f^{-1}(25) = ? \leftrightarrow f(?) = 25$

$f(x) = 25$

$3x^3 + 1 = 25$

$3x^3 = 24$

$x^3 = 8$

$x = 2$

$f'(x) = 9x^2$

$f'(2) = 9(4)$

$= 36$

Example 6: The table below gives values of the differentiable functions f and f' at selected values of x. Let $g(x) = f^{-1}(x)$.

a.) Find the value of $g'(1)$

$g'(1) = (f^{-1})'(1)$

$= \frac{1}{f'(f^{-1}(1))}$

$= \frac{1}{f'(2)} = -\frac{1}{2}$

$f^{-1}(1) = ?$

$f(?) = 1$

$f^{-1}(1) = 2$

x	$f(x)$	$f'(x)$
1	3	−3
2	1	−2
3	−5	−5
4	0	−6

b.) Write an equation for the line tangent to g at $x = 1$.

$x=1$ $g(1) = f^{-1}(1) = 2$

$g'(1) = \frac{-1}{2}$

$y - 2 = \frac{-1}{2}(x-1)$